

Reading Assignment 1 - AI-Assisted Reading Notes

Article Identification

- Levy Lisboa Pan, 2502010502
- Article: “Three insurtech trends to watch in 2020”
- Author: Harry Menear
- Publication: FinTech Magazine
- Date: May 2020
- Topic: InsurTech, digital transformation, AI, drones, wearables, IoT, and insurance disruption
- Article source: <https://fintechmagazine.com/crypto/three-insurtech-trends-watch-2020>

Article Summary

The article explains how the insurance industry is being reshaped by technology. Insurance is traditionally one of the older, more conservative areas of finance. It is usually slow to adopt new tools because it depends on risk management, regulation, large pools of capital, and consumer trust. The article argues that this slow-moving structure is being challenged by InsurTech firms that use AI, data analytics, machine learning, drones, connected devices, and app-based customer experiences.

The first trend discussed is chatbots and automation. The article presents chatbots as a way for insurers to improve customer experience, reduce processing time, answer questions, personalize policies, and process claims. Lemonade is the main example. Its app-based bots can produce quotes, handle customer interactions, and process certain claims automatically. The article highlights the famous Lemonade case in which a claim was processed in only a few seconds after fraud checks.

The second trend is drones. The article argues that drones can help insurers collect information after natural disasters, floods, fires, or other events where physical access may be dangerous or slow. Drones can inspect damaged areas, collect live video, improve underwriting and claims estimates, and reduce the risk and cost of human inspections. DropIn Inc. is used as the main example.

The third trend is wearables and IoT data. The article explains that insurance depends heavily on data collection and risk assessment. Wearables, vehicles, workplace sensors, and other connected devices can produce real-time behavioral and environmental data. This can help insurers price risk more dynamically, encourage healthier or safer behavior, and process claims with more information. Examples include John Hancock’s use of fitness-tracking data and MakuSafe’s workplace-safety wearable platform.

The overall argument is that insurance companies need to decide whether they want to be the disruptors or the disrupted. Digital transformation can make insurance faster, more personalized, more data-driven, and more efficient. However, the article also leaves room for important concerns, including privacy, data access, algorithmic bias, regulation, market concentration, and the limits of automation in sensitive financial services.

AI-Assisted Questions And Answers

Question 1: What is the main argument of the article, and what are the three InsurTech trends it identifies?

The main argument of the article is that insurance is being transformed by digital technology in the same general way that banking, payments, and investing have been transformed by FinTech. Insurance has historically been slow to change because it is capital-intensive, heavily regulated, and dominated by established firms. However, the article argues that a new generation of InsurTech companies is using technology to make insurance faster, more personalized, and more data-driven.

The article identifies three specific trends.

The first trend is AI chatbots and automation. Chatbots allow insurers to interact with customers through apps and online interfaces without requiring a human agent for every task. They can answer common questions, collect customer information, generate quotes, personalize policy options, and help process claims. The article uses Lemonade as the main example. Lemonade's model depends heavily on app-based automation and AI bots. This matters because traditional insurance processes are often slow and bureaucratic. If a chatbot can handle routine questions or claims, the insurer can reduce costs and improve customer experience.

The second trend is drones for damage assessment. The article explains that after natural disasters, insurers need to inspect damage quickly. This can be difficult if roads are blocked, buildings are unsafe, or the affected area is very large. Drones can collect images and video from places that are dangerous or difficult for humans to access. The example in the article is DropIn Inc., which provides live video inspections to support underwriting and claims handling. This trend matters because it can reduce inspection costs, reduce human risk, improve damage estimates, and speed up claim payments.

The third trend is wearables and IoT data. The article argues that data is central to how insurers understand risk. Wearable devices, vehicles, workplace sensors, and connected devices can provide continuous data about health, behavior, environment, and safety. This can help insurers move away from pricing based only on historical averages and toward more individualized risk assessment. John Hancock and MakuSafe are examples. John Hancock is discussed in relation to fitness tracking, while MakuSafe is discussed in relation to workplace safety and workers' compensation.

Overall, the article is not only about three isolated technologies. These technologies represent a broader movement toward automation, real-time data collection, and personalized insurance. The article's strongest point is that insurance companies cannot treat digital transformation as optional. They must either adopt these technologies carefully or risk losing ground to firms that can serve customers faster and with better data.

Sources: FinTech Magazine article.

Question 2: Looking back from today, were chatbots, drones, and wearables good predictions, or should the article have chosen broader categories?

With hindsight, the article's predictions were generally reasonable, but they were also somewhat narrow. Chatbots, drones, and wearables were not random choices. Each one represented a real technological direction in insurance. However, the deeper trend was broader than the three individual tools.

Chatbots were a good prediction because AI-assisted customer service, automated claims intake, and app-based policy management continued to become important after 2020. The limitation is that "chatbots" is too narrow as a category. The more important trend is AI in insurance operations. That includes customer support, document processing, claim triage, fraud detection, underwriting support, pricing support, internal knowledge search, and agent assistance. In other words, chatbots are only the visible customer-facing part of a much larger AI transformation.

Drones were also a reasonable prediction, especially for claims and damage assessment. They are useful when insurers need to inspect property after floods, hurricanes, fires, or other disasters. Drones can reduce the time and danger involved in physical inspections. However, drones are probably not as central to the entire insurance industry as AI and data analytics. They are very useful in specific lines of business, especially property insurance, catastrophe claims, and infrastructure-related claims, but they do not transform every insurance process.

Wearables were a strong prediction because they connect directly to real-time risk measurement. Insurance is fundamentally about estimating and pricing risk, so devices that generate behavioral and health data can be extremely valuable. The limitation is that the article could have framed this more broadly as IoT-based and data-driven insurance. Wearables are only one part of a larger ecosystem that includes connected cars, smart homes, industrial sensors, telematics, workplace devices, and health platforms.

My interpretation is that the author was trying to make concrete predictions rather than abstract ones. It is easy now to say that the article should have predicted "AI" or "data analytics" more broadly, but broad predictions are less specific. The article chose visible technologies that were already emerging in 2020. In that sense, it made a useful prediction. The better criticism is not that the article was wrong, but that the article did not fully explain how these specific technologies would become part of larger business systems.

The most accurate way to read the article is therefore: chatbots, drones, and wearables were early examples of larger changes in automation, remote inspection, and data-driven risk assessment. The article was directionally correct, but a stronger version would have connected each tool to a broader InsurTech category.

Sources: FinTech Magazine article.

Question 3: What are the main advantages and risks of using wearable and IoT data in insurance?

Wearable and IoT data can be valuable for insurance because they create more detailed and more current information about risk. Traditional insurance pricing often relies on broad categories, historical averages, demographic variables, and self-reported information. Wearables and connected devices can create more direct signals. For example, fitness trackers may provide data about activity levels, sleep, or health behavior. Workplace sensors may detect hazardous conditions. Vehicle telematics can measure driving behavior. Smart-home devices can detect leaks, smoke, or security risks.

The first advantage is better risk assessment. If an insurer has more accurate data, it may be able to price policies more precisely. This could benefit lower-risk customers, because they may receive lower premiums. It can also benefit insurers, because better data can reduce adverse selection and improve underwriting accuracy.

The second advantage is prevention. Insurance normally pays after a loss occurs, but IoT data can help prevent losses before they happen. For example, workplace sensors can warn about dangerous conditions before an injury occurs. Smart-home sensors can detect a water leak before it becomes a major property claim. Wearables can encourage healthier behavior. This changes insurance from only compensation after damage into a more active risk-management service.

The third advantage is claims efficiency. If a device records relevant data before or during an event, it may be easier to verify what happened. That can reduce fraud, reduce disputes, and accelerate payment. For policyholders, this can mean faster service. For insurers, this can mean lower claims-handling costs.

However, the risks are serious.

The first risk is privacy. Wearables and IoT devices can collect very sensitive information about health, movement, work conditions, home life, and behavior. Insurance customers may not fully understand what they are sharing or how it will be used. Consent can also become questionable if customers feel they must share data to receive affordable coverage.

The second risk is platform access. From my own experience trying to work with wearable data in a website project, the issue is not only privacy. There is also a practical technical problem. Wearable ecosystems are often closed. Data may be controlled by Apple, Fitbit, Garmin, Google, or other platform owners. External developers may not easily access the

data. Large insurers may be able to negotiate partnerships with these platforms, but smaller firms and independent developers may face much higher barriers.

The third risk is discrimination. If insurers use behavioral or health data to price coverage, the result may be more personalized, but it may also become more exclusionary. Customers with poorer health, risky jobs, disabilities, or limited ability to change their behavior could be penalized. Even if the model does not directly use protected characteristics, proxy variables may reproduce unfair outcomes.

The fourth risk is data quality and interpretation. Device data can be incomplete, inaccurate, or misleading. A wearable may fail to record activity correctly. A workplace sensor may not capture the full context of a hazard. A model may interpret behavior incorrectly. If insurers rely too heavily on this data, customers may face wrong prices, wrong risk classifications, or unfair claim decisions.

The fifth risk is cybersecurity. More data collection creates more valuable targets for attackers. Insurance data is already sensitive, and adding health, location, workplace, and device data increases the damage that could result from a breach.

For these reasons, wearable and IoT insurance should not be treated only as a technical improvement. It is also a governance problem. The technology can improve pricing, claims, and prevention, but only if insurers handle consent, privacy, data access, cybersecurity, bias testing, and human oversight responsibly.

Sources: FinTech Magazine article; EU AI Act Annex III; NAIC AI Model Bulletin.

Question 4: AI seems to have become one of the most important technologies in insurance. What is the current regulatory situation around AI in insurance? Is AI still mostly unregulated, or are there already specific rules? Could future regulation slow down or damage the growth of AI in InsurTech, or is the technology already too deeply adopted for regulators to seriously restrict it?

The idea that AI in insurance is “not regulated” is only partly true. There is not always one single law called “AI insurance regulation”, especially in the United States. However, AI in insurance is already constrained by existing insurance law, anti-discrimination law, data protection law, consumer protection rules, unfair trade practice rules, and newer AI-specific frameworks.

The better conclusion is that AI in insurance is not banned, but it is increasingly regulated when it affects high-impact decisions such as pricing, underwriting, eligibility, claims handling, risk scoring, and consumer treatment.

In the European Union, the EU AI Act directly matters for insurance. Annex III classifies AI systems used for risk assessment and pricing for natural persons in life and health insurance as high-risk AI systems. This does not mean insurers cannot use AI. It means the systems must meet stronger governance requirements. These include risk management,

data governance, technical documentation, transparency, record-keeping, human oversight, accuracy, robustness, and cybersecurity. The point of the EU approach is not to stop AI use completely, but to make high-impact AI systems auditable and accountable.

In the United States, the regulatory situation is more fragmented because insurance is mainly regulated at the state level. There is no single national U.S. AI-insurance law equivalent to the EU AI Act. However, U.S. regulators are clearly moving toward more oversight. The NAIC Model Bulletin on AI use by insurers sets expectations around fairness, accountability, compliance with state laws, transparency, and safe, secure, fair, and robust AI systems. The bulletin also makes clear that insurers using AI must still comply with applicable federal and state laws.

New York is an important example. New York DFS Insurance Circular Letter No. 7, issued in July 2024, addresses the use of artificial intelligence systems and external consumer data in insurance underwriting and pricing. The circular letter identifies DFS expectations for insurers that use AI, external consumer data, or predictive models in underwriting and pricing insurance policies and annuity contracts.

Colorado is also important, although the exact legal framework changed recently. The original Colorado AI Act created broad obligations around high-risk AI systems and algorithmic discrimination. In May 2026, Colorado moved to SB26-189, a newer automated decision-making technology framework. Even with this change, the direction is still clear: Colorado is trying to regulate automated systems that influence consequential decisions, including insurance. Separately, Colorado insurance regulation requires life insurers using external consumer data, algorithms, and predictive models to maintain governance and risk-management frameworks designed to detect and remediate unfair discrimination.

Regulation will probably not destroy AI growth in insurance. A more realistic result is that regulation will divide AI uses into lower-risk and higher-risk categories. Lower-risk uses will probably continue to grow quickly. These include internal document summarization, customer service support, claims triage, fraud flagging, agent assistance, and back-office automation. These uses can improve efficiency without fully determining a customer's access to coverage or price.

Higher-risk uses will face more scrutiny. These include pricing, underwriting, eligibility decisions, claims denial, health and life insurance risk scoring, and models that use sensitive variables or proxy variables. In these areas, regulators are likely to require documentation, bias testing, explainability, governance, auditability, and human oversight.

It is not too late for regulators to intervene. Insurance is already a heavily regulated industry, and regulators do not need to ban AI to influence its development. They can require insurers to document models, test for unfair discrimination, explain decisions, maintain human oversight, audit third-party vendors, and prove compliance. These obligations can strongly shape which AI systems are commercially viable.

Regulation may also affect startups and incumbents differently. Large insurers usually have legal teams, compliance departments, actuarial teams, model-risk-management processes, and money to absorb compliance costs. Smaller InsurTech startups may struggle more because regulatory compliance increases fixed costs. This creates a paradox: rules designed to make AI safer may also strengthen large incumbents because they are better able to comply.

My final judgment is that AI in insurance is no longer an unregulated experiment, especially when it affects pricing, underwriting, claims, eligibility, or consumer treatment. Regulation will probably not stop AI adoption, because the technology is already too useful and too deeply integrated into insurance operations. However, regulation will shape how AI is used. The strongest future AI use cases will be explainable enough, auditable, compliant with anti-discrimination rules, supported by human oversight, and useful without fully replacing human judgment.

My critical interpretation is that the 2020 article was right that AI would become important, but it did not seriously discuss regulation. With hindsight, that is a major missing issue. In insurance, AI is not just a productivity tool. It can affect access to financial protection, prices, and claim outcomes. For that reason, regulation will likely be one of the main forces shaping InsurTech after 2026.

Sources: EU AI Act Annex III; NAIC Model Bulletin on AI in insurance; New York DFS Insurance Circular Letter No. 7; Colorado SB26-189; Colorado Regulation 10-1-1.

Question 5: Has AI changed the competitive landscape of the insurance industry? More specifically, did AI and recent InsurTech advances allow smaller insurance companies to gain significant market share from traditional large insurers, or does the market still look mostly dominated by the same big players as in 2020?

AI has clearly changed how insurance companies operate, but it has not yet radically changed who dominates the insurance market. The market still appears to be mainly controlled by large incumbents. Smaller AI-native InsurTech firms have grown and influenced the industry, but they have not taken enough market share to seriously displace the largest traditional insurers.

The best evidence is market concentration. According to NAIC property and casualty market-share data, in 2025 the top 10 U.S. P&C groups had a cumulative market share of 47.87%, and the top 25 had 65.71%. The total market in that NAIC report was about \$1.11 trillion in direct written premiums. This means that nearly half of the market was still held by the top 10 groups, and about two-thirds was held by the top 25. That is not a market where small technology-native insurers have replaced incumbents.

Some InsurTech companies have grown significantly. Root, for example, reported that in 2024 it achieved its first profitable year, with net income of about \$31 million. It reported

414,862 policies in force and about \$1.3 billion in gross premiums written. This shows that a technology-driven insurer can scale and improve performance.

Lemonade also shows strong AI adoption. In its 2024 annual report, Lemonade stated that its claims bot, AI Jim, took the first notice of loss without human intervention 96% of the time, and that roughly 55% of claims were automated as of December 31, 2024. In its Q4 2025 shareholder letter, Lemonade reported \$1.24 billion in in-force premium and 2,984,513 customers. These are meaningful figures and show that AI-native insurers can build real businesses.

However, these companies are still small compared with the whole P&C market. If the U.S. P&C market has around \$1.11 trillion in direct written premiums, then an AI-native insurer with around \$1 billion to \$1.5 billion in premiums is still a small player at the industry level. It may be important as an innovator, but it is not yet large enough to change the overall market structure.

AI also does not help only startups. Large insurers can use AI too. Incumbents are adopting AI for underwriting, claims processing, fraud detection, customer support, pricing, risk modeling, document processing, and internal productivity. In some ways, AI may actually strengthen large insurers because they already have the key resources needed to deploy it at scale.

Large insurers have several advantages:

- more capital;
- more historical claims and underwriting data;
- stronger brands;
- existing customers;
- regulatory and compliance departments;
- actuarial teams;
- distribution networks;
- reinsurance relationships;
- ability to absorb technology failures and compliance costs.

This means that AI does not automatically level the playing field. Startups may be more agile, but incumbents have more data, money, and regulatory capacity. In insurance, those advantages matter a lot.

InsurTech may therefore be changing the industry more as a technology layer than as a market-share revolution. Many AI-focused InsurTech firms do not necessarily replace insurers. They sell tools to insurers, automate parts of the value chain, improve claims workflows, support underwriting, provide risk data, or help with fraud detection. This still matters, but it is a different type of disruption. It changes how insurance is produced more than it changes who controls the market.

Investment data supports this interpretation. Gallagher Re reported that AI-focused InsurTech firms received a very large share of InsurTech funding in Q1 2026, with AI-centered companies raising about \$1.55 billion and receiving 95.2% of global InsurTech venture funding. This shows that investors believe AI is central to the future of insurance. However, funding concentration is not the same thing as market-share disruption. A startup can raise money and still not become a dominant insurer.

My final judgment is that AI has changed the insurance landscape mostly at the operational level, not yet at the market-share level. The market still looks dominated by large insurers, while smaller AI-native insurers and InsurTech startups grow around them, compete in specific niches, or sell technology into their operations.

A good critical conclusion is that AI did not destroy the traditional insurance hierarchy between 2020 and 2026. Instead, it became a tool used by both startups and incumbents. Some smaller InsurTech firms grew quickly, but the market-share data still show strong continuity. The biggest change is not that small insurers replaced big ones, but that every insurer is now under pressure to become more data-driven and automated.

Sources: NAIC 2025 Property and Casualty Market Share report; Root Q4 2024 shareholder letter; Lemonade 2024 Annual Report; Lemonade Q4 2025 shareholder letter; Gallagher Re Q1 2026 Global InsurTech Report.

Personal Comments And Critical Reflection

My first impression is that, although the article could have framed some trends more broadly, its predictions were still reasonable. With hindsight, it is easy to say that the author should have focused on broader categories such as AI in insurance operations rather than only chatbots, or IoT-based insurance rather than only wearables. However, this criticism should be treated carefully. The purpose of the article was not to perfectly predict the future of the insurance industry, but to identify concrete technologies that were becoming relevant at the time.

In this sense, I think the article made a good prediction. Chatbots, drones, and wearables were not random examples. They were specific technologies that represented a larger movement toward automation, data collection, and digital transformation in insurance. The fact that some of them later became part of broader categories does not make the article wrong. It only shows that the technologies evolved and became integrated into larger business processes.

One point that interested me in particular was the use of wearables. I have personally tried to work with wearable data in a website project, and I noticed that the issue is not only privacy. Privacy and user consent are clearly important, but there is also a technical and platform-access problem. Wearable ecosystems are often closed, and it is not always easy for external developers to access the data. For a large insurance company, this may be easier because it has more bargaining power and may be able to negotiate with

companies such as Apple, Fitbit, Garmin, or other device providers. However, for smaller companies or independent developers, access to this type of data can be a major barrier.

The market-structure question also made me think. The evidence suggests that, despite technological innovation, the insurance industry is still dominated by large incumbents. This is not surprising. Insurance is a heavily regulated and risk-sensitive industry. Large insurers cannot easily experiment with unproven technologies because mistakes could harm customers, create legal problems, and generate very large financial losses. In that sense, slow adoption is not necessarily a weakness. It may actually be desirable, because insurance affects people's lives, homes, health, and financial protection.

At the same time, this creates an interesting question: if large players remain dominant and smaller InsurTech startups rarely become major insurers, why do startups continue entering the industry? My interpretation is that startups still have incentives to innovate because they can capture value before the technology becomes fully standardized. They may also become acquisition targets, sell technology to incumbents, or specialize in parts of the insurance value chain rather than trying to become full insurance companies. Therefore, the role of startups may not always be to replace large insurers. In many cases, their role may be to test new technologies, prove that they work, and then allow larger companies to adopt them later.

This also changed how I interpret the slow transformation of insurance. At first, slow adoption may seem like a sign of inefficiency. However, after thinking about the risks, I see it more as a structural feature of the industry. Insurance companies should not adopt every new technology immediately just because it is innovative. If a chatbot, AI model, drone inspection, or wearable-data system makes wrong decisions, the consequences can be serious. For this reason, a slower and more cautious adoption process may be better for consumers and for the stability of the industry.

The chatbot trend is also interesting because I have built chatbots for clients and used them in services such as banking apps. My experience is that chatbots are useful, but only for specific tasks. They work well when the question is general and when the answer can be retrieved from a clear knowledge base. For example, a banking chatbot can explain how to send money from the app or how to activate a card feature. However, many chatbots are still not deeply personalized. If I ask whether a specific feature is active on my own account, the chatbot usually cannot answer. Technically, this information probably exists in the bank's internal systems, but privacy, security, and regulatory concerns make this type of integration more difficult.

This suggests that the main value of chatbots is not full replacement of human workers. Instead, their value is in handling repetitive questions, guiding users through simple processes, and reducing the workload for human agents. Human support is still necessary, especially when the issue is sensitive, complex, or related to personal financial data. Even if AI becomes much more accurate, it will probably need human supervision in insurance and banking because these sectors require trust, accountability, and clear responsibility.

Overall, I think the article was directionally correct. It identified real technologies that continued to matter after 2020. However, the deeper lesson is that InsurTech transformation is not only about adopting new tools. It is about balancing innovation with risk, regulation, consumer trust, data access, and market structure. The most important insight I gained is that insurance is not slow to change only because companies are conservative. It is slow to change because the cost of technological failure can be very high.

Source List

- Menear, H. (2020). “Three insurtech trends to watch in 2020.” FinTech Magazine. <https://fintechmagazine.com/crypto/three-insurtech-trends-watch-2020>
- European Commission AI Act Service Desk. Annex III, high-risk AI systems. <https://ai-act-service-desk.ec.europa.eu/en/ai-act/annex-3>
- National Association of Insurance Commissioners. “The Use of Artificial Intelligence Systems in Insurance”, Model Bulletin. https://content.naic.org/sites/default/files/inline-files/2023-12-4%20Model%20Bulletin_Adopted_0.pdf
- New York Department of Financial Services. Insurance Circular Letter No. 7 (2024), “Use of Artificial Intelligence Systems and External Consumer Data and Information Sources in Insurance Underwriting and Pricing.” <https://www.dfs.ny.gov/industry-guidance/circular-letters/cl2024-07>
- Colorado General Assembly. SB26-189, “Automated Decision-Making Technology.” <https://leg.colorado.gov/bills/SB26-189>
- Colorado Regulation 10-1-1, governance and risk management framework for external consumer data, algorithms, and predictive models, via Cornell Legal Information Institute. <https://www.law.cornell.edu/regulations/colorado/Article-3-CCR-702-10-1-1-5>
- National Association of Insurance Commissioners. 2025 Property and Casualty Market Share report. <https://content.naic.org/sites/default/files/research-actuarial-property-casualty-market-share.pdf>
- Root, Inc. Q4 2024 shareholder letter filed with SEC. <https://www.sec.gov/Archives/edgar/data/1788882/000162828025008125/q42024-shareholderletter.htm>
- Lemonade, Inc. 2024 Annual Report. https://s24.q4cdn.com/139015699/files/doc_financials/2024/ar/LMND-2024-Annual-Report-Final-4-23-25.pdf
- Lemonade, Inc. Q4 2025 shareholder letter. <https://www.lemonade.com/investor-relations-bo/wp-content/uploads/docs/LMND-Shareholder-Letter-Q4-25.pdf>
- Gallagher Re. Q1 2026 Global InsurTech Report. <https://www.ajg.com/gallagherre/-/media/files/gallagher/gallagherre/news-and-insights/2026/may/global-insurtech-report-2026-q1-ai-digital-risks.pdf>